Physics 12 Winter 2026
Week 5 Discussion

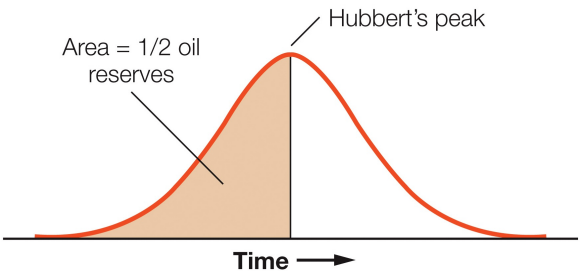
TA: Jiashu Han
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1. **Summary of Topics from Last Week**

- *From the lectures*
  - ◇ Human metabolism (chemical to internal)
    - ∗ Basal metabolic rate is 80 W.
  - ◇ Efficiency
    - ∗ Efficiency = (useful energy output) / (total energy input).
    - ∗ Thermal energy has better quality at higher temperature.
  - ◇ Cogeneration
    - ∗ Using captured waste heat
  - ◇ Fossil fuels (types, chemical composition/reactions, formation, usage, common units)
    - ∗ CO_2 from buring fossil fuels - 70% anthropogenic greenhouse gas emissions (2022).
  - ◇ Carbohydrate
    - ∗ Less energy released due to some oxidation already happened in the molecule.
  - ◇ Consumption of fossil fuels (demand/supply), Hubbert's curve (oil production rate)
    - ∗ You should know the distinction between discovered/proven/unproven reserves.
  - ◇ Carbon offset / credit
  - ◇ Coal (mining process, environmental and health impacts)
  - ◇ Unconventional fossil fuel sources: Fracking, tar sands
  - ◇ Air pollution (sulfur, nitrogen, etc.) and treatment
    - ∗ Buoyancy
    - ∗ Types of pollutants: gases, particulates (solid), aerosols (liquid droplets)
    - ∗ Types of treatments: cyclone, electrostatic separator/precipitator
- *From the textbook* (Wolfson 4th edition)

- ◇ Fossil fuels: origin, types, composition (Chapter 5.1-2)
- ◇ Extraction and Usage of fossil fuels: types of engines (Chapter 5.3 - 5.4)
- ◇ Fossil fuel resources (Chapter 5.5 - 5.6)
- ◇ Fossil fuel-related pollution, treatment (Chapter 6)

2. **Brain-warmer: Hubbert's Peak.** What's the quantity described in the vertical axis of this plot?



A. The rate of resource extraction

B. The growth in the resource extraction rate

C. Cumulative amount of resources extracted

D. Amount of remaining resources

E. Cost of extraction resources

Answer: A.

3. **Flying is dirty business**[1]**.** Professor travels a lot to Geneva, Switzerland for research on the Higgs boson. From SAN, he flies 6000 miles on a Boeing 777-300ER which burns 7 metric Tons of jet fuel per hour at a cruising speed of 550 Miles per hour. (Take off and landing consumes lot more fuel but for this long-distance flight, it can be folded in). 1 gram of jet fuel burnt produces 3.16 gram of CO_2 (like most hydrocarbons). The Professor occupies one of the 450 seats on the plane.

   How many Tons of CO_2 emission should he be held responsible for

   (a) For just this flight?

[1]From this week's homework

Solution: The total amount of jet fuel burned for the whole flight would be

$$6000 \text{ mi} \times \frac{1 \text{ h}}{550 \text{ mi}} \times \frac{7 \text{ t}}{1 \text{ h}} = 76.4 \text{ t} \tag{1}$$

We can then convert this to CO_2 emission using the ratio between jet fuel and CO_2. We further divide that number by the number of seats on the plane (assume it's full) to get the Professor's CO_2 contribution:

$$76.4 \text{ t jet fuel} \times \frac{3.16 \text{ g } CO_2}{1 \text{ g jet fuel}} \times \frac{1}{450/\text{passenger}} = 0.54 \text{ t/passenger} \tag{2}$$

(b) For 10 roundtrips to GVA per year?

Solution: For 10 round trips to GVA, we multiply the above number by 20 to get 10.7 tons of CO_2 emission per year for the Professor.

The Professor does not "do" carbon-credits. Instead, every year, the solar panels on his house generate 16 MWh of electrical energy to the SDGE electric grid, corresponding to 11.65 Tons of CO_2 equivalent. In balance, is he a CO_2 polluter?

Solution: Since the Professor's solar panels generate more energy from zero-emission energy sources than the equivalent CO_2 emissions he's responsible for, he is not a CO_2 polluter. Let's give him a shout out!

4. **Gas-fired power plant**[2]**.** Here's the analog of Example 5.2 for a natural gas power plant: Consider a gas-fired power plant that is rated at 600 MWe and is 44% efficient. Natural gas is supplied via pipeline. What rate of gas flow, in cubic meters per day, is required?

   Solution: The thermal power produced by the gas-fired power plant is

$$P_{\text{th}} = \frac{P_{\text{e}}}{e} = \frac{600 \times 10^6 \text{ W}}{0.44} = 1.36 \times 10^9 \text{ W} \tag{3}$$

According to Table 3.3, the energy content per volume of natural gas is

$$\frac{\text{energy}}{\text{volume}} = \frac{30 \times 10^3 \text{ W} \cdot \text{h}}{100 \text{ ft}^3} \left(\frac{3 \text{ ft}}{0.914 \text{ m}} \right)^3 = 1.06 \times 10^4 \text{ W} \cdot \text{h/m}^3 \tag{4}$$

[2] Wolfson (4th edition) Chapter 5 Exercise 2

The flow rate of natural gas required to provide the thermal power calculated above is

$$\text{flow rate} = \frac{\text{volume}}{\text{time}} = \frac{\text{energy/time}}{\text{energy/volume}} = \frac{1.36 \times 10^9 \text{ W}}{1.06 \times 10^4 \text{ W} \cdot \text{h/m}^3} = 1.28 \times 10^5 \text{ m}^3/\text{h} \tag{5}$$

Convert this to cubic meters per day, we get 3.1×10^6 m^3/day.

5. **CO_2 from gasoline[3].**

Estimate the amount of CO_2 released in burning 1 gallon of gasoline. Give your answer in both kilograms and pounds. The density of gasoline is about 730 kg/m^3, and gasoline is about 84% carbon by weight.

Solution: mass of C = 0.84 mass of gas

mass of gas = density×volume

volume = 1 gallon

$$\text{mass of C} = 0.84 \times 730 \frac{\text{kg}}{\text{m}^3} \times 1 \text{ gal} \left(\frac{3.83 \text{ L}}{\text{gal}}\right) \left(\frac{10^3 \text{ cm}^3}{\text{L}}\right) \left(\frac{\text{m}}{10^2 \text{ cm}}\right)^3 = 2.3 \text{ kg} \tag{6}$$

This converts to 5.2 lbs of carbon emitted per gallon of gas.

$$\text{mass of } CO_2 = \frac{44}{12} \text{ mass of C} = 8.5 \text{ kg} \left(\frac{2.2 \text{ lb}}{\text{kg}}\right) = 19 \text{ lbs } CO_2 \text{ per gallon of gas} \tag{7}$$

6. **Liquefied natural gas[4].**

A large LNG tanker carries 138,000 m^3 of liquefied natural gas. (a) Use Table 3.3 and the fact that LNG is 600 times denser than the gaseous state to estimate the total energy content of the tanker's cargo. (b) How long would that natural gas last if it were used to satisfy the total U.S. energy-consumption rate of about 3 TW?

Solution: The energy content of the tanker's cargo is

$$\text{energy}_{\text{LNG}} = 600 \left(\frac{\text{energy}}{\text{volume}}\right)_{\text{NG}} (\text{volume}) \tag{8}$$

where

$$\left(\frac{\text{energy}}{\text{volume}}\right)_{\text{NG}} \left(\frac{1000 \text{ Btu}}{\text{ft}^3}\right) \left(\frac{1054 \text{ J}}{\text{Btu}}\right) \left(\frac{3.28 \text{ ft}}{\text{m}}\right)^3 = 3.7 \times 10^7 \text{ J/m}^3 \tag{9}$$

[3] Wolfson (4th edition) Chapter 6 Exercise 1

[4] Wolfson (4th edition) Chapter 6 Exercise 4

(a) Using the equations above, we get

$$\text{energy}_{\text{LNG}} = 600\left(3.7\times10^{7}\ \text{J/m}^3\right)(1.38\times10^{5}\ \text{m}^3) = 3\times10^{15}\ \text{J} = 3\ \text{PJ} \quad (10)$$

(b)

$$\text{time} = \frac{\text{energy}}{\text{power}} = \frac{3\times10^{15}\ \text{J}}{10^{12}\ \text{J/s}} = 3000\ \text{s} \approx 1\ \text{h} \quad (11)$$

7. **Waste heat at nuclear power plant**[5]**.**

Before it shut down in 2014, the Vermont Yankee nuclear power plant drew cooling water from the Connecticut River in the winter at the rate of 360,000 gallons per minute. The plant produced 650 MW of electric power with an efficiency of 34%. (a) At what rate did the plant discharge waste heat? (b) By how much did the temperature of the cooling water rise as a result of the waste heat it absorbed? You may want to review Section 4.4 on specific heat before tackling this exercise.

Solution:

$$P_{\text{th}} = \frac{P_{\text{e}}}{e} = \frac{650\ \text{MW}}{0.34} = 1.9\times10^{3}\ \text{MW} \quad (12)$$

Most of this power was wasted as heat: $(1-e) = 66\%$.

(a)

$$\text{rate of heat flow} = \frac{dQ}{dt} = 66\%\times P_{\text{th}} = 0.66\times1.9\times10^{3}\ \text{MW} = 1.3\ \text{GW} \quad (13)$$

(b) Assume all the waste heat went into the water. Then the water temperature rose by a ΔT implied by Equation 4.3: $Q = mc\Delta T$ or $\frac{dQ}{dt} = \frac{dm}{dt}c\Delta T$, where the rate of mass flow is

$$\frac{dm}{dt} = (\text{density})\times\frac{dV}{dt} = \left(\frac{1\ \text{kg}}{\text{L}}\right)\times\frac{360000\ \text{gal}}{\text{min}}\left(\frac{\text{min}}{60\ \text{s}}\right)\left(\frac{3.8\ \text{L}}{\text{gal}}\right) = 2.3\times10^{4}\ \text{kg/s} \quad (14)$$

So, the temperature change is

$$\Delta T = \frac{dQ/dt}{c\ dm/dt} = \frac{1.3\times10^{9}\ \text{J/s}}{(4184\ \text{J/kg/K})(2.3\times10^{4}\ \text{kg/s})} = 13\ \text{K} \quad (15)$$

[5] Wolfson (4th edition) Chapter 6 Exercise 14